

Use of habitat features, edges and harvest treatments by spruce grouse in subalpine forest

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Abstract

Winter use of habitat features, edges and alternative forest harvest types by spruce grouse (*Falcapennis canadensis*) was measured at the Sicamous Creek Silvicultural Systems site in high-elevation forest in south-central British Columbia, Canada. Habitat relationships and edge effects were tested at five other validation sites. Winter use by grouse was indicated by droppings in habitat plots or edge transects conducted shortly after snowmelt. Assumptions of using droppings to indicate grouse occurrence were assessed. Grouse high-use areas (100 m-radius) were distinguished from areas available overall by higher densities but lower basal areas of fir and spruce trees, presence of surficial rock and sparse shrub cover. Plots of 0.01 ha used by grouse within high-use areas were distinguished from unused plots by these same habitat features, plus greater canopy cover and more relatively short trees. Knolls were preferred topography types, as predicted by this habitat model. Use of forest was reduced within 10 m of recently harvested openings of 0.1, 1 and 10 ha, and within 20 m of older large openings. Use increased with proximity to open wetlands until 20 m from the wetland, paralleling predicted habitat quality, then declined nearer to the edge despite high predicted quality. Percent decline in winter occurrence of grouse was similar to the percent timber removed (33%) in harvest treatments with 0.1, 1 and 10 ha openings, but a more pronounced decline occurred in uniform partial cuts, where uniform thinning lowered predicted habitat quality. Recommendations to maintain winter habitat for spruce grouse in managed high-elevation forests include maintaining forest patches on knolls or other poor growing sites and around wetlands, ensuring some retained area >10 m from harvested and natural edges, and avoiding widespread use of uniform partial cutting.

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1. Introduction

Maintaining native species throughout their ranges has become an important objective in forest management, with particular concern expressed in many areas for forest-dwelling species that avoid the early-seral forest stages created extensively by industrial forestry

(FEMAT, 1993; Haila, 1994; CSP, 1995; Lindenmayer et al., 2000). For many species dwelling in mature forest, effects of habitat loss and fragmentation can be reduced by maintaining features in the matrix of harvested forest that allow continued use by species associated with older forests (Franklin et al., 1986; Wilcove, 1989; Franklin, 1989). Species most likely to benefit from knowledgeable management of the harvested forest are those that are sensitive to habitat loss and fragmentation, but which are not entirely restricted to undisturbed old forest.

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Alternatives to large clear-cuts, such as individual-tree selection (ITS) partial cuts, green-tree retention, variable-retention or small patch-cut systems, are being recommended and used to try to maintain forest-dependent species in managed forest (Franklin, 1989; CSP, 1995). These alternative silvicultural systems may produce a matrix that is less hostile to many species associated with older forests. However, because these silvicultural systems disperse harvesting, they must also be applied to a larger area than clear-cuts to obtain the same volume of timber, and they can create extensive edges, resulting in loss of forest interior conditions (Franklin and Forman, 1987). Determining the value of these alternative systems for forest-dwelling species requires empirical measurement of how organisms respond to the trade-off between retaining forest structure in the residual stand, but affecting larger areas and increasing the amount of edge. Additionally, effective matrix management that minimizes conflict with timber production requires knowledge of which habitat features are essential to retain for the species of concern.

Spruce grouse (*Falci pennis canadensis*) inhabit boreal and high-elevation forests and are of potential concern to managers because they are negatively affected by forest harvesting, which is occurring at an increasing rate in these forest types (Braun et al., 1994). Although spruce grouse use openings such as wetlands, roadsides, and some harvested areas for breeding and foraging in summer (Pietz and Tester, 1982; Boag and Schroeder, 1992), they are restricted to conifer trees for 7–9 months of winter, when their diet consists of conifer needles (Pendergast and Boag, 1970). Spruce grouse have been extirpated in areas where agricultural development has permanently reduced conifer forest at the edge of their range (Fritz, 1979), and they are uncommon in small, isolated patches of conifer forest (Whitcomb et al., 1996). A long history of intensive forest management is threatening the persistence of populations of the ecologically-similar capercaillie (*Tetrao urogallus*) in many places in Europe (Rolstad, 1989; de Franceschi and Bottazzo, 1991; Swenson and Angelstam, 1993).

Spruce grouse can be abundant in young stands of lodgepole pine (*Pinus contorta*) or jack pine (*P. banksiana*) that have regenerated following wildfires (Schroeder and Boag, 1991). Ensuring successful regeneration and appropriate rotation lengths may

be adequate to sustain spruce grouse in pine-dominated forests (Boag and Schroeder, 1992). The suitability of young stands is less certain in some high-elevation forests where pine species do not occur, and where slow regeneration combined with deep snowpacks can result in a lack of substantial conifer cover during winter for many years following harvesting. Spruce grouse prefer particular habitat features, such as dense patches with greater canopy cover (Allan, 1985; Boag and Schroeder, 1992), suggesting that retention of important stand elements may be a primary management option to mitigate effects of harvesting in forest types lacking pine regeneration.

Effects of the edges of harvested openings and of different silvicultural systems are poorly known for spruce grouse. Edges of wetlands adjacent to forests are preferred for nesting and summer use in Minnesota, but this response changes in winter when grouse are using conifer trees (Pietz and Tester, 1982), and may differ for openings created by harvesting. Siberian spruce grouse (*F. falci pennis*) prefer areas near swamp edges in winter, but are expected to decline in extensively harvested landscapes (Andreev, 1990). Capercaillie nest on clear-cut edges (Storch, 1991) and show little response to edges of natural openings in winter, but suffer greater mortality in areas fragmented by clear-cuts (Gjerde and Wegge, 1989; Gjerde, 1991; Wegge et al., 1992). Scattered small openings are a preferred harvest system for capercaillie (Rolstad and Wegge, 1987). The same system has been suggested for spruce grouse (Boag and Schroeder, 1992), but is untested for this species.

This study was designed to provide information required to improve the suitability of managed high-elevation forests for spruce grouse, as part of a large project examining many resource and ecosystem values (Vyse, 1999). Objectives were: (1) to develop predictive models of winter habitat use by spruce grouse that would identify important habitat elements, which could be used to help predict effects of alternative forest management options on spruce grouse in winter; (2) to measure winter use by grouse of forest near edges of natural and harvested openings, to help in design of the size and shape of openings and retained patches; and (3) to evaluate directly the effects of alternative silvicultural treatments of winter use by spruce grouse. The study focused on winter use by spruce grouse, because forest management most

directly affects the conifer trees that the species relies on for the 7–9 months of winter conditions in high-elevation forests. A comprehensive strategy to maintain spruce grouse in managed forests must also include provision of breeding habitat and dispersal opportunities, because these are clearly also important to the persistence of spruce grouse populations (e.g. Boag et al., 1979; Boag and Schroeder, 1987).

2. Methods

Table 1 summarizes the methods and analyses used to meet the above objectives.

2.1. Main study site and validation sites

The study was conducted at the Sicamous Creek Silvicultural Systems site, 7 km south-east of Sicamous, British Columbia, Canada (50°50'N, 118°50'W). The site is on a moderate (3–20°) north-facing slope, 1530–1830 m elevation, in the wet cold subzone of the Engelmann spruce (*Picea engelmannii*)—subalpine fir (*Abies lasiocarpa*) zone (ESSFwc2; Lloyd et al., 1990). Subalpine fir and Engelmann spruce are the only tree species in this subzone, with 82% fir and 18% spruce among canopy trees at Sicamous Creek. The oldest trees across the site are approximately 350 years old (Parish et al., 1999). Understory consists of dense shrubs (*Rhododendron albiflorum*, *Menziesia ferruginea*, and

Vaccinium species) or herbaceous plants (*Rubus pedatus*, *Valeriana sitchensis*, *Streptopus roseus*, *Tiarella trifoliata*, *Gymnocarpium dryopteris*) and an almost ubiquitous moss layer. Soil moisture ranges from sub-xeric in shallow soils on elevated, rocky topography to hygric, open wetlands in poorly drained areas.

The site was divided into 15 contiguous study units of 30 ha each (Fig. 1). Each unit was assigned one of five treatments, in a randomized design blocked by elevation: (1) single 10 ha clear-cut with surrounding 20 ha unharvested area; (2) array of nine 1 ha patch-cuts with 100 m wide unharvested leave strips; (3) array of 60 0.1 ha patch-cuts with 30 m wide unharvested leave strips; (4) ITS partial cuts with 20% uniform removal of the two tree species across the size distribution, and removal of all trees along skid trails; and (5) uncut control. Each of the four harvest treatments removed 33% of the trees in a study unit. Harvesting occurred in mid-winter 1994–1995. Home ranges of spruce grouse are typically less than the size of these treatment units (<24 ha; Boag and Schroeder, 1992), but seasonal movements can be longer. This study looks mainly at finer-scale selection of habitat features within the individual's range. Grouse are not hunted at the Sicamous Creek site, due to restricted access.

The generality of habitat models and edge relationships from Sicamous Creek were tested at five validation sites located 10–90 km from the main study site. All validation sites were in forest >140 years old in the

Table 1
Summary of analyses of winter use by spruce grouse

Objective 1: To develop predictive models of habitat use
Compare openings versus forest (no use of openings; omit openings subsequently)
Compare topography types (strong preference for knolls; omit this rare type in subsequent model development)
Discriminant function analysis at broader scale: compare systematic plots (available habitat) with Sicamous GCPs ^a (3–4 ha “high-use areas”)
Discriminant function analysis at finer-scale: compare used versus unused plots within GCPs, separately for Sicamous Creek and validation sites
Test Sicamous Creek model versus data from validation site, and vice versa
Test Sicamous Creek model with data from knolls
Objective 2: To measure use of forest near edges of natural and harvested openings
Direct measurement with transects in forest parallel to cutblock edges
Occurrence in Sicamous Creek GCPs versus distance from edge of openings
Compare with predicted occurrence based on discriminant function habitat model
Objective 3: To evaluate the effects of alternative silvicultural treatments
Direct comparison of occurrence in preferred habitat types in different treatments

^a GCP: grouse-centered plots (set of plots centered on area with high grouse activity; see text).

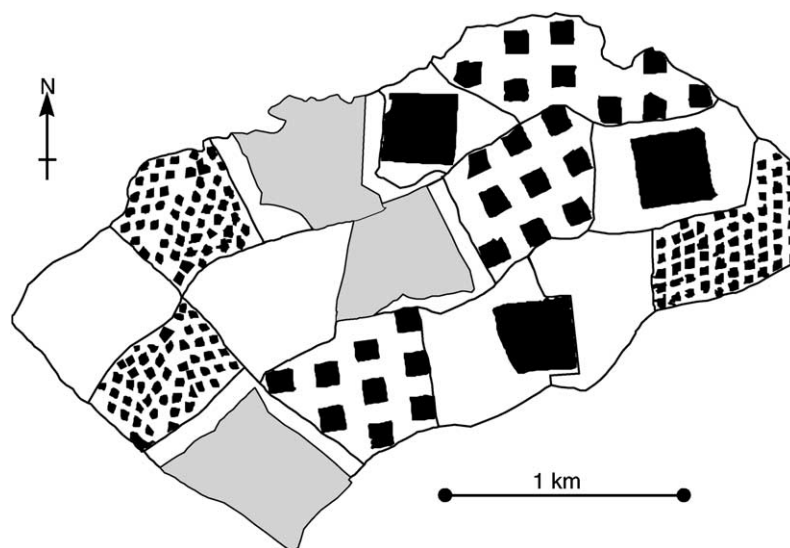


Fig. 1. The Sicamous Creek Silvicultural Systems study area. Black areas are cutblocks of 0.1, 1, and 10 ha; gray areas are ITS blocks; white is uncut forest.

ESSFwc2 subzone, with 54–90% subalpine fir trees and the remainder Engelmann spruce. Slopes were $<20^\circ$ with variable topography and aspect. Clear-cuts of 20–60 ha, from 3 to 20 years old, were scattered across the sites. These openings had sparse conifer regeneration <2 m tall. Grouse may be hunted in these areas, but access is difficult.

2.2. Habitat models—field plots

Use of habitat features by spruce grouse was measured in 1995 and 1996 using circular plots of 5.65 m radius (0.01 ha). In each plot, observers recorded: percent canopy cover; percent shrub cover; species, diameter at breast height (DBH) and height of each tree ≥ 7.5 cm DBH; slope; topography in one of six classes (flat or uniform slope, rolling slope, elevated knoll, side or base of knoll, ridge or bench, and gully); the presence of bedrock or large boulders at the surface; distance to the edge of the nearest cutblock; and distance to the nearest open wetland. Trees and canopy were measured because these affect grouse occurrence (Allan, 1985; Boag and Schroeder, 1992). Shrub cover, rock, topography and slope were measured as possible indicators of site conditions that could affect the trees that grouse use. Canopy and shrub covers were estimated visually, with all observers trained by

one person. Tree heights were estimated, and periodically standardized to heights of trees measured with a clinometer. Sampling was conducted immediately after snowmelt, prior to the emergence of leaves on shrubs or herbaceous plants, generally during early to late June. Observers carefully examined the ground and logs in the 0.01 ha plot for grouse droppings (feces composed of the dry fibrous remains of conifer needles in a distinctive cylindrical shape, 1–3 cm long, 5 mm in diameter). The presence or absence of droppings indicated the use or lack of use of the plot area by spruce grouse during the previous winter (see tests of this assumption later).

These plots to measure occurrence of grouse droppings and habitat features (“habitat plots”) used two sampling designs (Fig. 2): (1) systematic plots were conducted every 100 m along parallel transects covering the study site at 250 m intervals; (2) “grouse-centered plots” (GCPs) were centered on a location with high grouse activity. Early morning surveys were used during the spruce grouse display season prior to snowmelt to locate areas of spruce grouse activity for the centers of GCPs. Observers walked the parallel transects and perpendicular transects forming a grid at 250 m spacing, recording the locations of any displaying male grouse, females heard or seen, tracks in the snow or concentrations of grouse droppings. Transect

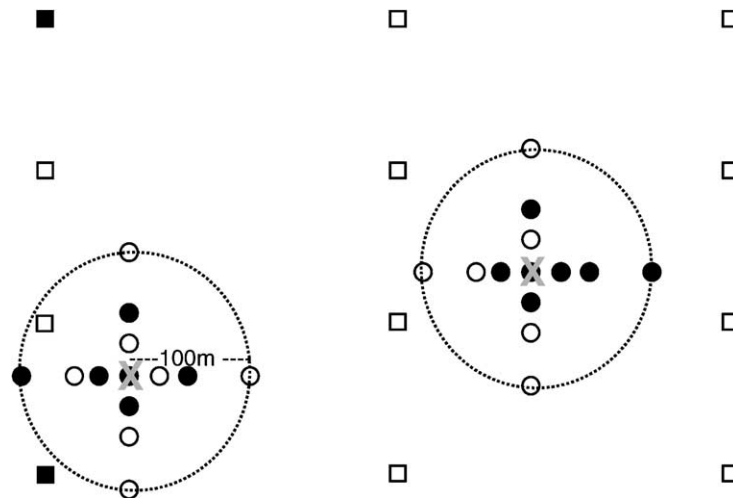


Fig. 2. Sampling design for systematic and grouse-centered plots. Squares: systematic plots, circles: grouse-centered plots (2 sets of 13 plots shown within 100 m (dashed line) of known areas of high activity by grouse). Solid symbols indicate plots with grouse sign.

surveys were repeated three or four times per year, with birds or their sign often recorded in the same location on each repetition. A 5.65 m radius habitat plot was surveyed at the center of each of these high activity areas, and 25, 50 and 100 m away to the north, east, south, and west. These 13 plots comprised one set of GCPs. Individual plots within a set of GCPs were occasionally not sampled if they fell on roads or in standing water or were inaccessible because of cliffs. The central plots of sets of GCPs were separated by >200 m. Transect surveys and GCPs at areas of high grouse activity were conducted in 1995 and 1996.

This sampling design allowed comparisons at two scales of areas used by grouse versus unused areas. (“Unused” here means “with no evidence of use”). Habitat features in all systematic plots, as a sample of available habitat in the study area, were compared to all GCP plots, to indicate how habitat features in 3–4 ha areas where grouse occurred (100 m radius) differed from the study area overall (Fig. 2; squares versus circles). Within GCPs, individual plots with grouse sign were compared to those without sign, to indicate local habitat use choices within 3–4 ha areas where grouse were known to occur (Fig. 2; solid versus open circles). Used and unused systematic plots were not compared because of low rates of grouse occurrence in systematic plots.

In 1997, 19 sets of GCPs were surveyed at the validation sites. The same methods were used, with

sets of plots centered where spruce grouse or accumulations of their sign were observed at the different sites.

2.3. Habitat models—analysis

Plots that fell in harvested openings (except ITS units) or in open wetlands never contained grouse sign in the spring surveys and were excluded from analyses. Proportions of plots with grouse droppings in each of the six topography types were compared using a χ^2 -test, separately for the Sicamous GCPs and the validation GCPs. Grouse use of knolls, which are relatively rare topographic features, was considerably higher than use of non-knoll plots. Plots on knolls were not used in subsequent development of habitat models, because these preferred sites are readily identified without the need for any further habitat modeling.

Habitat modeling sought to identify habitat elements that could distinguish areas in the forest used by grouse from unused areas. Nine variables were used from the habitat plots: slope, canopy cover, shrub cover, presence or absence of surface bedrock, number of live subalpine fir and Engelmann spruce and their basal areas, and number of “relatively short” trees. “Relatively short” trees are those that are shorter than expected from the observed relationship between tree height and DBH of trees in systematic plots. This relationship was nearly identical for the two species, and the combined relationship was used: expected

height = $-24.61 + 32.87 \log_{10} \text{DBH}$ ($r^2 = 0.76$, $n = 741$). This variable was included because Mueller (1993) found that spruce grouse prefer trees with low height:diameter ratios. All variables were transformed using $\log_{10}(x + 1)$ because this improved the normality of the distribution.

Linear discriminant functions based on these nine habitat variables were used to distinguish: (1) all grouse-centered plots at Sicamous Creek from all systematic plots at Sicamous Creek; (2) used from unused plots within GCPs at Sicamous Creek; and (3) used from unused plots within GCPs at the validation sites. The predictive ability of the Sicamous Creek GCP model was tested using the independent data from the validation sites, and vice versa. The Sicamous Creek GCP model, based on non-knoll plots, was also used to classify plots on knolls at Sicamous Creek, as a further test of the model's predictive ability and to address the question of whether the observed preference for knolls could be predicted by habitat features used off of knolls. Discriminant functions were calculated using Systat 7.0.

The 13 plots in a set of GCPs may not be fully independent samples because of spatial auto-correlation. The method of Clifford et al. (1989) was therefore applied to the GCP data sets to calculate the effective number of independent samples. Spatial autocorrelation resulted in an effective sample size of 328.5 at Sicamous Creek, reduced from the raw sample size of 415 GCP plots (excluding knolls and open areas), and an effective sample size of 169.8 at validation sites, close to the raw sample size of 173. The statistical significance of the discriminant functions were evaluated using degrees of freedom based on these effective sample sizes (Clifford et al., 1989). Numbers of observations for χ^2 -analysis of topography types were reduced proportionally to sum to the effective sample size. These procedures reduced the risk of obtaining spurious results due to spatial auto-correlation (Clifford et al., 1989), although tests with independent data were the primary means of evaluating the habitat models.

2.4. Edge effects—response to edges of openings

To assess use by spruce grouse of forest adjacent to harvested openings, transects were surveyed parallel to the edges of cutblocks at various distances into the

forest: 5 and 10 m from 0.1 ha openings; 5, 10, 20 and 50 m from 1 ha openings; and 5, 10, 20, 50 and 100 m from 10 ha openings. Each transect was the entire length of the edge of the square opening being surveyed (32 m for 0.1 ha openings, 100 m for 1 ha openings and 316 m for 10 ha openings), except that parts of the transect were not used if they were closer to a road or another opening than to the adjacent opening being surveyed. Observers recorded the number of grouse droppings observed within 1 m of the transect. Edge transects at Sicamous Creek were conducted in 1995, 1996 and 2001. Transects 100 m into the forest were not conducted in 2001 because of short edge effects observed in previous years. Edge transects of 200 m length were conducted in 1997 adjacent to the larger clear-cuts at the validation sites, at 5, 10, 20, 50 and 100 m into the forest.

Droppings per 100 m length was calculated for each transect, as an index of grouse use at the different distances from the edge. This density was then transformed using $\log(x + 1)$, because this improved the normality of the distribution and reduced the influence of high values in individual transects on the mean. The mean and standard error of this index were calculated at each distance from each size of opening, using the three replicates of each treatment (or five separate validation sites) as the sample unit.

Edge responses at Sicamous Creek were examined in a second way, by plotting the observed presence or absence of grouse in the GCPs against distance to harvested openings or, separately, to natural wetland openings. A fourth-order polynomial regression line was used to indicate the form of the relationship between percent occurrence and edge distance. A fourth-order equation was used because it was flexible enough to indicate the main form of the edge relationship. Expected occurrence based on habitat features alone was plotted in the same way, using the presence or absence of grouse in each plot predicted by the discriminant function habitat model. Only non-knoll plots were used for this analysis.

2.5. Treatment effects—use of preferred habitat in different harvest treatments

Habitat use plots showed a much higher occurrence of grouse sign on knolls and in forest 20–35 m from open wetlands. To assess whether different harvest

treatments affected the use of these preferred habitat types, 5.65 m radius plots were conducted on each knoll or within 20–35 m of each open wetland in all 15 replicate units at Sicamous Creek, recording the number of grouse droppings found in a careful search of the plot. Three 5.65 m radius plots spaced 15 m apart were conducted on each knoll or adjacent to each open wetland, except on small knolls where only one or two plots would fit. The proportional occurrence of grouse sign in all plots in each replicate was calculated, with a mean and standard error for each treatment based on the three replicates.

2.6. *Testing assumptions of droppings as an index of habitat use*

Relying on droppings to indicate winter use by spruce grouse involves a number of assumptions, including that other grouse species are not present, and that droppings are not produced at different rates in different areas. No other species of grouse have been recorded at Sicamous Creek in 9 years of nearly continuous field work. The low-quality, high-volume diet of spruce grouse in winter results in a regular production of feces (cf. capercaillie, Klaus et al., 1986 cited in Gjerde, 1991), with little likelihood of different rates of fecal production in the limited range of different habitats available in winter (Ratti et al., 1984). The method also assumes that grouse droppings are equally likely to be detected in different sites or positions relative to edges, and that they do not show different inter-annual persistence in different types. These two assumptions were tested directly. Droppings were collected from the snow in late winter, and approximately 20 droppings were distributed on the snow at each of 30 sites, 10 sites in the forest away from openings, 10 sites 2–5 m into the forest adjacent to harvested openings and 10 sites in harvested openings. Sites were marked and revisited at snow melt (when plot surveys and edge transects began), 4 weeks later (when surveys and transects were finished), and at the end of the summer. The observer looked for droppings with the same effort used in the plots or edge transects, to assess whether the droppings would have been observed in a normal survey.

Translocated droppings were detected at all 20 sites in forest and edge locations at snowmelt and one month later. Droppings were not visible at 2 of 10 sites in

openings, probably because of loss due to surface flow of melt water or disintegration caused by freezing and thawing in spring. By late summer, droppings could still be found at most sites, but they had decayed from their characteristic cylindrical form, and were only readily visible at two forested and two edge sites, because they were generally covered by conifer needles or leaves of shrubs and herbs. These observations support the assumptions that droppings are equally visible in different forest types and edge positions during surveys immediately after snowmelt, and that very few, if any, droppings remain intact and visible enough to be counted in the following year's survey. Examination of four previously occupied sites where grouse disappeared confirmed the latter assumption, with evidence of only highly decayed droppings. The occurrence of grouse droppings in surveys conducted shortly after snowmelt is therefore taken to be a reliable indication of the occurrence of spruce grouse during the preceding winter.

3. Results

3.1. *Occurrence by topography type*

Grouse droppings were found in 19 of 165 systematic plots (11.5%) and in 204 of 503 individual plots in the GCPs (40.6%) at Sicamous Creek (including knolls and open areas). Only 3 of the 165 systematic plots (1.8%) fell on knolls, because these occupy only a small part of the study area, while 32 of 503 grouse-centered plots (6.4%) were on knolls. The proportion of GCPs with grouse droppings differed by topography type at Sicamous Creek ($\chi^2 = 21.4$, d.f. = 5, $P = 0.001$). Plots on knolls had a higher occurrence of grouse droppings than plots in other topography types, with plots on ridges or benches having intermediate occurrence (Fig. 3). Grouse sign occurred in 93 of 242 GCPs (38.4%) conducted at validation sites, with knolls also showing higher occurrence than other topography types (Fig. 3). Ridges and benches did not show intermediate levels of occurrence in the validation GCPs, but only 10 plots were in this topography type.

3.2. *Habitat models at two scales*

The habitat characteristics of all non-knoll GCPs differed significantly from the characteristics of

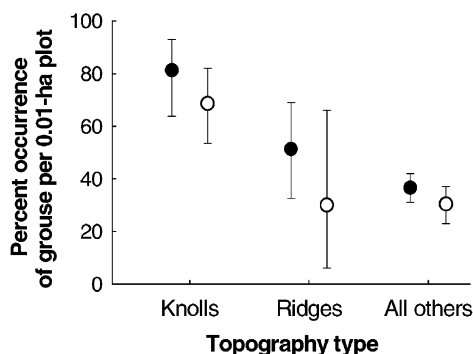


Fig. 3. Percent occurrence of grouse droppings in 0.01 ha plots in three topography types. Filled symbols: Sicamous Creek plots; open symbols: plots at validation sites. Error bars are binomial 95% confidence intervals.

systematic plots at Sicamous Creek (Wilks' $\lambda = 0.88$, approximate $F = 8.53$, d.f. = 9, 483, $P < 0.001$), demonstrating habitat selection at this larger scale. Grouse-centered plots had higher densities of spruce and subalpine fir, but lower basal areas of these species, especially spruce (Table 2). They also had less shrub cover and lesser slopes, and more commonly had rock exposed at the surface. The discriminant function identified 115 of 156 systematic

plots (73.7%) correctly and 254 of 415 grouse-centered plots (61.4%) correctly.

At the finer-scale, the discriminant functions to separate plots with and without grouse sign within GCPs (i.e. microhabitat use within areas generally used by grouse) were highly significant for both data sets (Sicamous Creek: Wilks' $\lambda = 0.823$, approximate $F = 9.64$, d.f. = 9.318, $P < 0.001$; validation sites: Wilks' $\lambda = 0.757$, approximate $F = 5.82$, d.f. = 9,159, $P < 0.001$). The standardized coefficients for both data sets indicated increased probability of grouse presence with increasing stem densities of spruce and subalpine fir trees, and with the presence of bedrock at the surface (Table 2). At Sicamous Creek, probability of grouse presence also increased with decreasing basal area of spruce and subalpine fir, despite the positive relationship with the densities of trees. More, smaller trees were therefore preferred. The validation sites did not show this negative relationship with basal area. Increasing canopy cover and increasing numbers of relatively short trees led to increased probability of grouse at Sicamous Creek, but the validation plots showed no relationships with these variables.

The discriminant functions for Sicamous Creek and validation sites GCPs correctly classified 67.2 and

Table 2
Discriminant function habitat models for spruce grouse at two scales

Variable ^a	High-use areas versus systematic		Used versus unused GCP plots			
	Coefficient ^b	Standardized	Sicamous Creek		Validation sites	
			Coefficient	Standardized	Coefficient	Standardized
Constant	0.24		-1.72		-3.25	
Slope	-0.64	-0.25	-0.13	-0.05	-0.07	-0.02
Canopy cover	0.23	0.09	1.63	0.54	-0.36	-0.08
Shrub cover	-0.87	-0.48	-0.23	-0.01	-0.28	-0.11
Rock	0.70	0.28	0.60	0.23	1.26	0.48
Density <i>B</i>	1.83	0.54	2.41	0.65	3.18	0.54
Density <i>S</i>	2.47	0.84	1.27	0.38	2.59	0.57
Basal area <i>B</i>	-0.29	-0.14	-2.33	-1.00	-0.19	-0.05
Basal area <i>S</i>	-0.68	-0.49	-0.52	-0.32	0.57	0.28
Short trees	-0.24	-0.08	1.08	0.32	0.25	0.06

^a Slope (°), canopy cover (%), shrub cover (%), rock = absence (0) or presence (1) of surficial rock, density *B* = number of live subalpine fir per 0.01 ha, density *S* = number of live spruce per 0.01 ha, basal area *B* = basal area of live subalpine fir (m²/ha), basal area *S* = basal area of live spruce (m²/ha), short trees = number of relatively short live trees (see text) per 0.01 ha plot. Only trees >7.5 cm DBH are counted. All variables are transformed as $\log_{10}(x + 1)$.

^b Discriminant function is calculated as the constant plus each of the variables multiplied by their coefficient. Values >0 indicate increased probability that the plot was in a high-use area, or used by grouse. Standardized coefficient is coefficient divided by the standard deviation of the variable, to compare the relative importance of individual variables.

71.1%, respectively, of the plots in their data sets. The models were less successful at predicting the presence or absence of grouse in the grouse-centered plots at the other sites, with overall success of 60.1 and 65.1%. The discriminant functions based on the habitat features in the non-knoll Sicamous Creek and validation site GCPs predicted that grouse would be present in 67.7 or 66% of the plots on knolls at their respective sites, and the predicted presence or absence was correct in 78.1 or 76% of the cases, respectively. Habitat models developed with non-knoll plots therefore did predict the observed preference for knolls.

3.3. Edge effects

Transect surveys adjacent to the three sizes of openings at Sicamous Creek shortly after harvesting (1995 and 1996) showed considerably lower densities of grouse droppings 5 m into the forest than at 10 m from the openings, while densities at 10 m and further from the edge were the same (Fig. 4a). Transects adjacent to larger and older openings at the validation sites showed a lower density of droppings 5–20 m

from edges compared to 50 or 100 m into the forest (Fig. 4b), although uncertainty was greater due to less sampling effort and more variable edges. Reduced use at 10 m into the forest was also seen in 2001 at Sicamous Creek adjacent to 10 ha openings.

The occurrence of grouse sign in Sicamous Creek grouse-centered plots was lower if the plot was within 10 m of a harvested opening than in more distant plots (Fig. 5a). The predicted occurrence of grouse, based on the discriminant function model, increased with proximity to open wetlands (Fig. 5b). The actual occurrence of grouse in the GCPs adjacent to wetlands followed this predicted pattern, except that observed occurrence declined at distances of <15 m from the wetland edge (Fig. 5b). The peak in observed occurrence was thus at 20–35 m from wetland edges, while maximum predicted occurrence based on habitat occurred at the wetland edge.

3.4. Harvest treatment effects

Grouse sign occurred in 55.8% of all plots on knolls or in areas 20–35 m from open wetlands in uncut

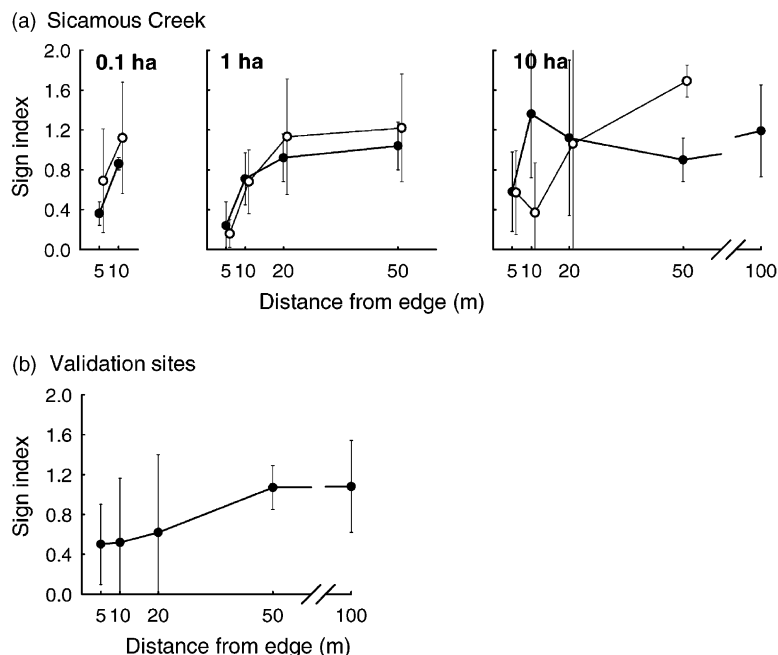


Fig. 4. Relative abundance of grouse droppings at different distances into the forest from harvested openings at Sicamous Creek (a) and validation sites (b). Filled circle: 1995, 1996 survey; open circle: 2001 survey. The three opening sizes at Sicamous Creek are shown separately. Error bars are 95% confidence intervals. Sign index = $\log_{10}((\text{number of droppings} + 1) \text{ per } 100 \text{ m of transect})$.

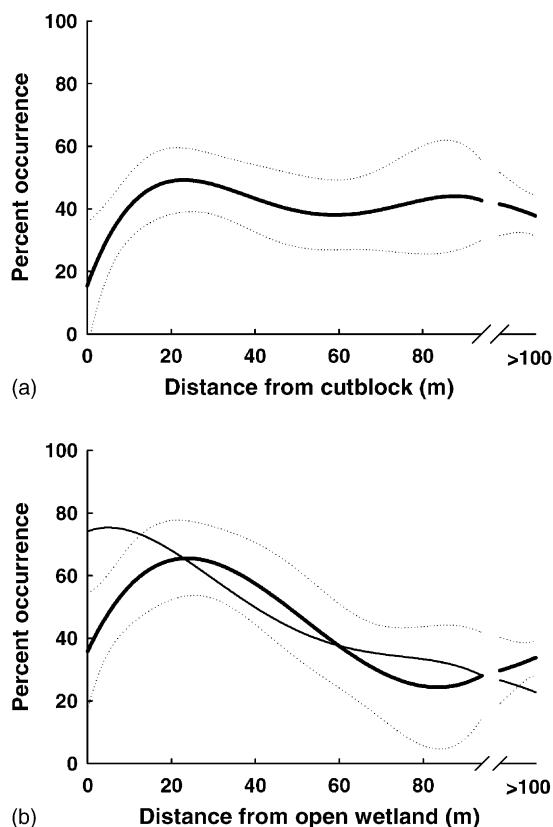


Fig. 5. Occurrence of grouse droppings in grouse-centered plots as a function of distance from cutblock (a) and distance from open wetland (b). Fourth-order polynomial lines (heavy line) and 95% confidence intervals (dotted lines) were fitted to the actual presence/absence data (data not shown). Distances >100 m from an edge were truncated to 100 m. Predicted occurrence, based on a fourth-order polynomial fitted to the presence or absence at each sample point predicted by the discriminant function model, is shown in (b) (thin line).

control blocks in 1996 and in 76.5% in 2001 (Fig. 6). Occurrence of grouse in the forested two-thirds of the 0.1 ha patch-cut array, 1 ha patch-cut array and 10 ha clear-cut treatments was nearly equal to the occurrence in the uncut control in both years. Because grouse sign was never observed in the openings that occupy one-third of these treatments, the overall effect of these treatments was therefore to lower occurrence of grouse by one-third, regardless of the size of the openings or leave strips (Fig. 6). The reduction in occurrence in the ITS treatment units was greater shortly after harvesting, with only 17.9% of plots in preferred habitat types having grouse sign (Fig. 6), a 68%

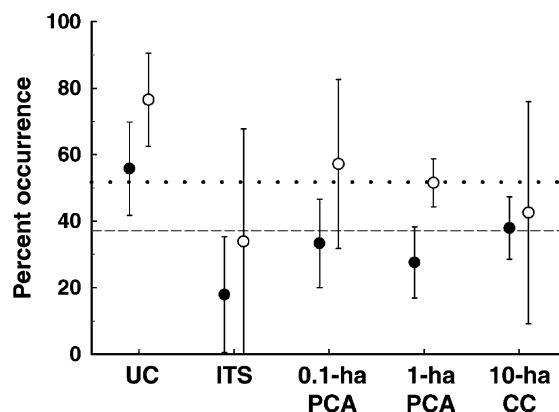


Fig. 6. Occurrence of grouse droppings in preferred habitat (on knolls or in forest 20–35 m from open wetlands) in five treatments: UC = uncut control; ITS = individual-tree selection partial cut; 0.1 ha PCA = 0.1 ha patch-cut arrays, 1 ha PCA = 1 ha patch-cut arrays; 10 ha CC = 10 ha clear-cut and surrounding leave strip. Filled circle: 1996 survey; open circle: 2001 survey. Error bars are 95% confidence intervals based on $n = 3$ replicate blocks. Horizontal lines are at 66.7% of the occurrence observed in controls (dashed: 1996, dotted: 2001).

reduction in occurrence compared to uncut controls. In 2001, ITS still had the lowest occurrence, but the difference from other harvested treatments was less pronounced.

4. Discussion

4.1. Habitat use by spruce grouse

The habitat models generated with discriminant functions had only moderate classification success and predictive ability with independent data. Imperfect ability to predict grouse presence or absence in individual 0.01 ha plots is not an impediment to using the models to inform forest management, because applied uses are never concerned about particular very small areas. Instead, the habitat models only need to predict changes in habitat quality over differently managed larger areas (e.g. stands). The utility of the discriminant function models is supported by the similarity of models from the Sicamous Creek and validation sites, their moderate independent predictive ability, and the fact that they correctly predicted increased use of knolls and increased occurrence with proximity to open wetlands (mitigated by a

negative edge effect independent of habitat). The habitat relationships therefore seem to have validity across a range of habitat types.

The habitat models showed non-random use of habitat features by spruce grouse at two scales, the locations of high-use areas (100 m radius) and the use of plots (5.65 m radius) within high-use areas. Several habitat features were selected at both scales at Sicamous Creek, including greater densities but lower basal areas of the two tree species, and greater occurrence of surficial rock. High-use areas were also characterized by lower shrub cover and lesser slopes than the habitat available overall. The relationship with shrubs is unlikely to be a direct one, because shrubs are generally completely covered in snow all winter at SC. Instead, the relationship is likely an indirect reflection of grouse choosing particular stand types, which are characterized by a lack of shrub cover. Canopy cover and the density of relatively short trees were positively related to use of plots within high-use areas. Preference for dense canopy is commonly reported for spruce grouse in winter (e.g. Allan, 1985; Boag and Schroeder, 1992). A preference for short trees has been reported as a correlate of selection for younger post-fire seral stages at a larger spatial scale in pine forests (Schroeder and Boag, 1991), but this differs from the selection reported here for relatively short trees (shorter than expected from their diameters) within old forest.

Several of these habitat features characterizing sites used by spruce grouse suggest a selection for areas with poor conditions for tree growth, including small, relatively short trees, and surficial rock. This observation is supported by the strong preference grouse show for knolls and areas near open wetlands, where shallow soils on rock outcrops or water tables near the surface can limit tree growth. Toxic secondary compounds that restrict herbivore feeding on conifer needles are particularly important to spruce grouse in winter (Bryant and Kuropat, 1980; Mueller, 1993). However, trees in poor growing sites tend to produce greater concentrations of these secondary compounds, especially if they are limited by nitrogen availability (Mattson, 1980; Herms and Mattson, 1992). Structural characteristics of trees in poor growing sites, including lower, denser canopies, are compatible with thermal strategies or predator avoidance (cf. hazel grouse, Swenson and Olsson, 1991; and blue grouse, Pekins

et al., 1991), and may explain the preference of spruce grouse for these sites. Intensive silviculture in regenerating managed stands tends to produce lower densities of large, relatively tall trees with higher, sparser canopies (Smith et al., 1997), which these habitat models would predict as poor grouse habitat. However, direct surveys in managed, mid-rotation spruce-fir forests are needed to test these predictions.

4.2. Edge effects

Unlike many operational cutblocks, the systematically located experimental openings at Sicamous Creek do not confound edge position with changes in stand type or topography. The short but strong negative edge effect adjacent to harvested openings immediately after harvesting is therefore an inherent effect of edge, and not due to changes in habitat quality near the edge. A similar inherent negative edge effect was also apparent adjacent to the natural openings of wetlands, where observed occurrence dropped below the predicted increasing trend in habitat quality near wetlands. The moderate extension of the negative edge effect around older operational openings and around large openings at Sicamous Creek after 6 years may be attributable to windthrow, which was extensive in some places 5–25 m into the forest adjacent to large openings (Huggard et al., 1999). The resulting opening of the canopy reduces predicted habitat quality for grouse. However, with equal area cut, even a 25 m unused edge buffer around a 10 ha opening would cause less reduction of total useable forest (17% of area of uncut leave strips affected) than a 5 m unused edge around 0.1 ha openings (36.5% of area of leave strips affected).

Two mechanisms could plausibly explain the short negative edge effect, avoidance of predators and avoidance of microclimatic extremes. Avian predators are the dominant cause of mortality for overwintering grouse (Ellison, 1974), but the main raptor present at Sicamous Creek in winter is the Northern Goshawk (*Accipiter gentilis*), which is believed to be adapted more for hunting in forests than in openings. Thus, predators should not directly cause grouse to avoid forest near openings, but there may still be a negative edge effect simply because grouse perceive a greater risk of predation near openings. More likely, frequent severe weather in winter, including the accumulation

of blowing snow, produces inhospitable conditions in the forest at the edges of openings. Although, some microclimatic edge effects are detectable >240 m into forests, windspeed decreases to less than half the speed in openings within 30 m of entering the forest (Chen et al., 1995; Kremsater and Bunnell, 1999). Field observations in winter at Sicamous Creek show an increase in snow loading on conifer branches near edges, but only immediately adjacent to openings. This extra snow loading, which can completely and persistently cover the foliage, would eliminate foraging opportunities for grouse. The area of the forest immediately adjacent to an opening would therefore have both reduced foraging opportunities and greater thermoregulatory costs due to higher winds, both effects that penetrate only short distances into the uncut stand.

4.3. *Effects of alternative silvicultural systems*

Differences in the winter occurrence of spruce grouse between treatments were assessed in this study using only preferred habitat types that could be identified independently of the harvest treatment (knolls and areas adjacent to wetlands), because this allowed an acceptable degree of power to detect treatment effects with the available sampling effort. Based on confidence intervals of the binomial distribution (Fig. 2 of Krebs, 1989), a 33% decline in occurrence in a treatment should be detected 95% of the time using 31 plots in preferred habitat (where occurrence of grouse in controls was 60%). In contrast, random or systematic plots (where occurrence of grouse in controls is only 10%) would have required 400 plots to detect the same percent decline 95% of the time. A limitation of surveying only preferred habitats is that small reductions in density may not be detected if the remaining individuals concentrate their activities more in the preferred habitats. On the other hand, random surveys would also risk not detecting reductions, unless large numbers of surveys could be conducted.

Occurrence of spruce grouse in preferred habitats in the patch-cut and clear-cut treatments dropped approximately in proportion to the percent of forest removed (33%; Fig. 6). A greater decrease was expected in the 0.1 ha patch-cut array units based on the observed negative edge effect, because 36.5% of the uncut forest in this treatment is within 5 m of a cutblock edge. However, even 6 years after harvest, use of the 0.1 ha

patch-cut arrays was equal to treatments with larger openings and larger leave strips. The unexpected lack of greater decline in 0.1 ha patch-cut array units may indicate that the preferred habitat types surveyed in the treatments are used even if they are immediately adjacent to small openings. The lack of difference between the harvest systems with openings of different sizes may also be due to chance with limited numbers of replicates, and should be tested at other sites.

The percent reduction in occurrence of grouse in the uniform ITS treatment shortly after harvest exceeded the percent of timber volume removed (68% reduction in occurrence versus 33% volume removal). Unlike patch-cut arrays, which retain intact leave strips with habitat features unaffected at the 0.01 ha scale (plot size), ITS affects all of the stand at a fine scale, reducing tree densities and canopy cover throughout. If a fixed volume of timber is to be removed from an area of previously uncut forest, use of a patch-cut or clear-cut system thus would have less immediate impact on occurrence of spruce grouse in winter than a uniform ITS system.

4.4. *Implications for maintaining spruce grouse winter habitat in managed high-elevation forests*

Studies demonstrating that spruce grouse use more open forest conditions in summer imply that partial cuts or small patch-cuts could maintain or improve the breeding habitat of spruce grouse (Hedberg, 1980; Boag and Schroeder, 1992). Some spruce grouse move from breeding to wintering habitat, but such movements are short (median distances of 0.8 km for females, 0.4 km for males; Schroeder, 1985), suggesting that suitable winter habitat must also be maintained near suitable breeding habitat. Maintaining suitable habitat throughout the managed matrix of the landscape will reduce the highly uncertain risks due to landscape-level fragmentation. Results of this study suggest some strategies for maintaining winter habitat for spruce grouse in managed high-elevation forests:

1. Elevated knolls, even those only 2 m higher than surrounding topography, should be candidate sites for reserves in harvested areas. Sites with rock at or near the surface should be particularly favored as reserves. In general, poorer growing sites with denser, but smaller and particularly shorter trees would be good

reserve patches for wintering grouse. Choice of such sites as reserves is compatible with some forestry objectives, because these are difficult growing sites and difficult sites to regenerate; however, windthrow problems may be greater on exposed sites with shallow soils (Ruel, 1995). Incorporating buffers of greater than 10 m width around reserves on knolls would alleviate the negative edge effect for grouse, and might reduce risk of windthrow.

2. Areas near wetlands should also be considered as reserves. Current regulations and standard practices guiding riparian management in British Columbia generally provide only buffers of limited width around wetlands (typically 10 m in the Sicamous Creek area). These narrow areas immediately adjacent to wetland openings may be good habitat for spruce grouse, but are unlikely to be used as they are within 10 m of edges in two directions. Patches of at least 40 m width are recommended adjacent to wetlands for spruce grouse winter habitat, because these allow approximately one-half the area to be >10 m from an edge. For spruce grouse, a 40 m riparian buffer strip around part of a wetland would be preferable to a 10 m strip around the entire wetland.
3. The short extent of the negative edge effect suggests that loss of forest interior conditions is unlikely to be a concern for spruce grouse winter habitat, unless abundant edges are created throughout the forest. However, arrays of small patch-cuts that created abundant edge did not seem to have a large negative effect on grouse. Large-scale fragmentation with very large clear-cuts may also threaten spruce grouse by preventing successful dispersal, but effects at this scale were not addressed in this study.
4. Uniform ITS systems should not be applied to extensive areas where spruce grouse winter habitat is a management concern, unless suitable reserves are incorporated. The partly conflicting results obtained for the 0.1 ha patch-cut arrays (i.e. negative edge effects, but no apparent increased impact of the edge-rich arrays of 0.1 ha patch-cuts) suggest caution is also warranted about extensive applications of small patch-cut systems. Systems using moderate-sized patch-cuts are more likely to provide suitable winter habitat for spruce grouse in the associated leave strips, without special reserves being required. The longer-term effects of these systems depends on

the unknown suitability of the regenerating stands when the leave strips are removed in subsequent harvest entries. Variable-retention systems could also be appropriate for spruce grouse, as these systems have the flexibility to leave some patches of trees that are not uniformly thinned, and can easily incorporate reserves on knolls and adjacent to wetlands.

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